SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

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FACULTY OF SCIENCE AND HUMANITIES

DEPARTMENT OF COMPUTER APPLICATIONS

PROJECT REPORT

on

DATA ANALYSIS USING PYTHON PROGRAMMING

"COVID-19 ANALYTICS"

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Abstract:

This project focuses on the design and development of a full stack pandemic surveillance dashboard aimed at analyzing global COVID19 timeseries data and presenting actionable public health insights through an interactive interface. In the wake of global health crises, organizations and researchers rely heavily on real-time analytics to monitor infection trends, assess healthcare outcomes, and enhance strategic response efforts. This project demonstrates how raw epidemiological data can be transformed into meaningful surveillance intelligence using modern data analytics and web technologies.

The system utilizes a comprehensive dataset derived from the Johns Hopkins University (JHU) CSSE report, which includes vital information such as confirmed cases, fatalities, and patient recoveries across multiple geographic regions. Data preprocessing is performed using Python libraries such as Pandas and NumPy to clean, transform (wide-to-long format), and standardize the dataset, including the calculation of critical metrics like mortality and recovery rates. The processed data is subsequently stored in a MySQL relational database, featuring optimized indexing for efficient timeseries querying and high-performance analysis.

A Flask based backend is implemented to provide robust RESTful APIs for dynamic data retrieval and analytics processing, further enhanced by a custom in-memory caching mechanism to ensure low latency responsiveness. The frontend dashboard, developed using HTML, CSS, Bootstrap, and Chart.js, delivers insights through sophisticated and interactive visualizations. Key analytics include daily infection trends, global mortality rankings, continent-wise distribution, country-to-country comparison tools, and an animated bar chart race for dynamic ranking visualization.

The project highlights the seamless integration of healthcare data analytics and modern web development, emphasizing scalability, performance optimization, and intuitive usability. It serves as a practical example of how surveillance systems can leverage data-driven architectures to gain critical insights into global health trends and pandemic progression.

Introduction:

The global scale of the COVID19 pandemic has led to the generation of unprecedented volumes of epidemiological and transactional health data. This data, when systematically analyzed, provides critical insights into infection vectors, recovery trends, and the efficacy of public health interventions. However, extracting meaningful intelligence from raw, multiregional datasets requires advanced data processing, statistical analysis, and dynamic visualization techniques.

Traditional reporting methods and static datasets are often insufficient for tracking the rapid and highly dynamic nature of pandemic progression. This creates a vital need for integrated data analytics systems that can process largescale timeseries data efficiently and present findings through intuitive, real-time interactive interfaces. The convergence of data science and web based dashboards is essential for effective pandemic surveillance and informed decision making.

This project addresses these critical challenges by developing a full stack COVID19 analytics platform tailored for global surveillance. The system seamlessly combines backend data engineering leveraging Python, Pandas, and MySQL with a high-performance frontend powered by Flask and Chart.js to create a responsive analytics experience. By integrating automated data cleaning pipelines with interactive visualizations, the project demonstrates a comprehensive approach to solving complex, real-world public health data problems.

The significance of this project lies in its ability to bridge the gap between fragmented raw data and actionable public health insights. It showcases how modern data analytics can be applied to monitor global health trends, identify mortality hotspots, and enhance the overall speed and accuracy of pandemic response strategies.

Project Explanation:

The project is designed as a complete data analytics pipeline that transforms raw pandemic timeseries data into meaningful public health insights through multiple stages, including data ingestion, preprocessing, storage, analysis, and visualization.

The dataset used in this project is the Johns Hopkins University (JHU) CSSE COVID19 report, containing attributes such as confirmed cases, deaths, recoveries, and geographic location information including country and region. This global timeseries dataset serves as the foundation for all analytical operations and trend forecasting.

The first stage involves data ingestion and preprocessing using the Pandas library within the ***prepare_dataset.py*** script. During this stage, raw CSV files are cleaned by handling missing or null values and converting date columns into a standardized datetime format. A critical transformation involves "melting" the wide-format data (where dates are columns) into a tidy long-format structure. The pipeline also calculates advanced metrics like ***mortality_rate*** and ***recovery_rate*** and filters the data for the Top 50 most affected countries to ensure analytical focus and system performance. ***NumPy*** is utilized to handle complex numerical computations and avoid division by zero errors.

Following preprocessing, the cleaned data is stored in a MySQL relational database. The database schema, defined in ***schema.sql***, is carefully designed to support efficient timeseries querying. It consists of the primary ***`covid_data`*** table for historical metrics and a ***country_info*** table for continent-level mapping. Both tables utilize strategic indexing on ***country*** and ***date*** fields, allowing for rapid aggregation and retrieval of largescale datasets.

The backend of the system is developed using ***Flask***, a lightweight Python web framework. It provides a suite of **RESTful APIs** that handle data requests from the frontend, such as ***`/api/dailytrend`*** and ***`/api/keyinsights`***. The analytics engine, implemented using **Pandas** on the server-side, performs dynamic grouping, cumulative aggregation, and peak finding operations. To ensure high responsiveness, a custom in-memory caching mechanism is implemented to store frequently accessed results, significantly reducing database overhead.

The frontend dashboard is built using HTML, CSS, Bootstrap, and JavaScript, with Chart.js used as the primary engine for data visualization. Users can apply dynamic filters to toggle between global overviews and specific country drilldowns. The system also includes advanced features such as a Dark/Light mode toggle, an animated Bar

Chart Race, and a Data Explorer for custom analysis, all providing updates without requiring page reloads.

Important highlights of the project:

1. **Full Stack Data Pipeline:** End-to-end transformation from raw JHU CSSE data to interactive BI.
2. **Advanced Preprocessing:** Comprehensive data cleaning, long format transformation, and automated rate calculations using Pandas.
3. **Relational Storage:** Optimized MySQL schema with strategic indexing for high-performance timeseries analysis.
4. **Responsive API Backend:** Flask driven REST API with custom caching for minimal latency.
5. **Interactive Visualizations:** 10+ dynamic chart types powered by Chart.js, including animated rankings and correlation plots.
6. **UX/UI:** Professional design system with glassmorphism effects, dark mode support, and seamless AJAX driven updates.

Project Architecture:

The architecture of the system is designed as a multistage data processing pipeline, where each component performs a specific role in transforming raw epidemiological data into meaningful public health insights. This layered architecture ensures modularity, scalability, and efficient handling of largescale timeseries data.

The process begins with the **JHU CSSE CSV** datasets, which act as the primary source of raw data. This dataset contains unprocessed global COVID19 records, including timeseries statistics for confirmed cases, fatalities, and recoveries. Since raw epidemiological datasets often contain missing values or fragmented regional entries, they require significant transformation before they can be effectively analyzed.

The next stage is data cleaning and preprocessing using Pandas. In this stage, the dataset is refined by handling null values and converting wideformat data into a standardized "long" formata process known as melting. This step also involves calculating critical performance indicators such as mortality and recovery rates at the record level. This preprocessing is crucial because the accuracy of pandemic trend analysis depends heavily on the integrity and structure of the input data.

After preprocessing, the cleaned data is stored in a **MySQL database**. The use of a relational database ensures structured storage and highperformance querying for timeseries analytics. The database schema is optimized with specific tables for timeseries data and static countrycontinent mappings. Strategic indexing on ***country*** and ***date*** fields allows for rapid execution of complex aggregations and crossregional comparisons.

The **Flask backend API** acts as an intermediary layer between the database and the user interface. It handles client requests, retrieves relevant data from the MySQL storage, and applies a custom caching layer to minimize server load. This separation of concerns improves system maintainability and allows for independent iteration of the data processing and visualization layers.

Within the backend, the **analytics engine** (implemented using **Pandas**) performs the core data analysis operations. This includes calculating ***global KPIs***, identifying ***peak infection dates***, generating ***monthly accumulation trends***, and performing correlation analysis between ***cases*** and ***deaths***. The analytics engine transforms raw database records into aggregated, trendresponsive insights.

Finally, the processed data is presented through the Chart.js dashboard, which serves as the visualization layer of the system. The dashboard renders data through a professional glassmorphism-based UI, featuring interactive line charts, animated bar races, and summary metric cards. Users can apply dynamic country filters and custom analysis parameters to update visualizations in realtime without requiring page reloads, enabling a seamless data exploration experience.

Project Architecture diagram:

JHU CSSE COVID-19 Dataset (CSV)

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Data Preprocessing (Pandas)

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MySQL Database

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Flask Backend API

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Analytics Engine (Pandas)

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Chart.js Dashboard

Project Folder Structure:

The project follows a modular folder structure that separates different components based on their functionality, making the system organized and easy to maintain.

```
covid19
├── run_project.py
├── requirements.txt
├── schema.sql
├── backend
│   ├── app.py
│   └── config.py
├── dataset
│   ├── confirmed.csv
│   ├── deaths.csv
│   ├── recovered.csv
│   └── clean_covid_data.csv
├── scripts
│   ├── prepare_dataset.py
│   └── load_data_mysql.py
├── static
│   ├── css
│   │   └── style.css
│   └── js
│       ├── dashboard.js
│       ├── global_overview.js
│       ├── country_analytics.js
│       ├── trends.js
│       └── reports.js
└── templates
    ├── index.html
    ├── global_overview.html
    ├── country_analytics.html
    ├── trends.html
    ├── reports.html
    └── settings.html
```

This structure begins with core files such as ***run_project.py***, which serves as the main entry point to validate the environment and launch the Flask application. The ***requirements.txt*** file lists all necessary Python dependencies, including **Flask, Pandas, and MySQL-Connector,** required to run the project. The system's blueprint is defined in ***schema.sql***, which contains the SQL definitions for building the relational database.

The ***backend/*** directory contains the core server-side logic. This includes ***app.py,*** which handles RESTful API endpoints and data aggregation, and ***config.py***, which manages essential configuration settings like MySQL database connection parameters.

The ***dataset/*** directory serves as the data repository, storing the raw JHU CSSE time-series CSVs (`***confirmed.csv***`, `***deaths.csv***`, `***recovered.csv***`) alongside the final `***clean_covid_data.csv***` generated by the pipeline.

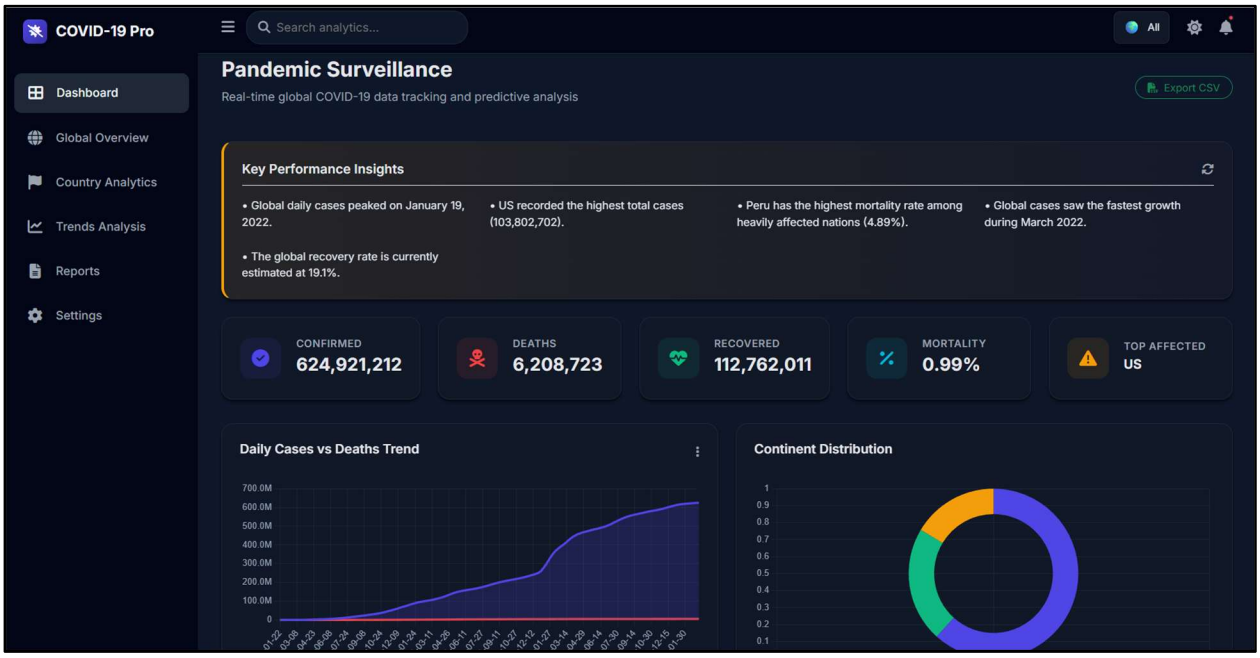
Automation and data engineering logic are housed in the ***scripts/*** directory, which contains specialized scripts for data cleaning ***prepare_dataset.py*** and automated database loading ***load_data_mysql.py***.

The ***static/*** directory holds all frontend assets, organized into subfolders for ***css/*** and ***js/*** (dynamic dashboard logic and Chart.js initialization). The ***templates/*** directory contains the HTML files, such as ***index.html*** and ***global_overview.html***, used to render the professional user interface.

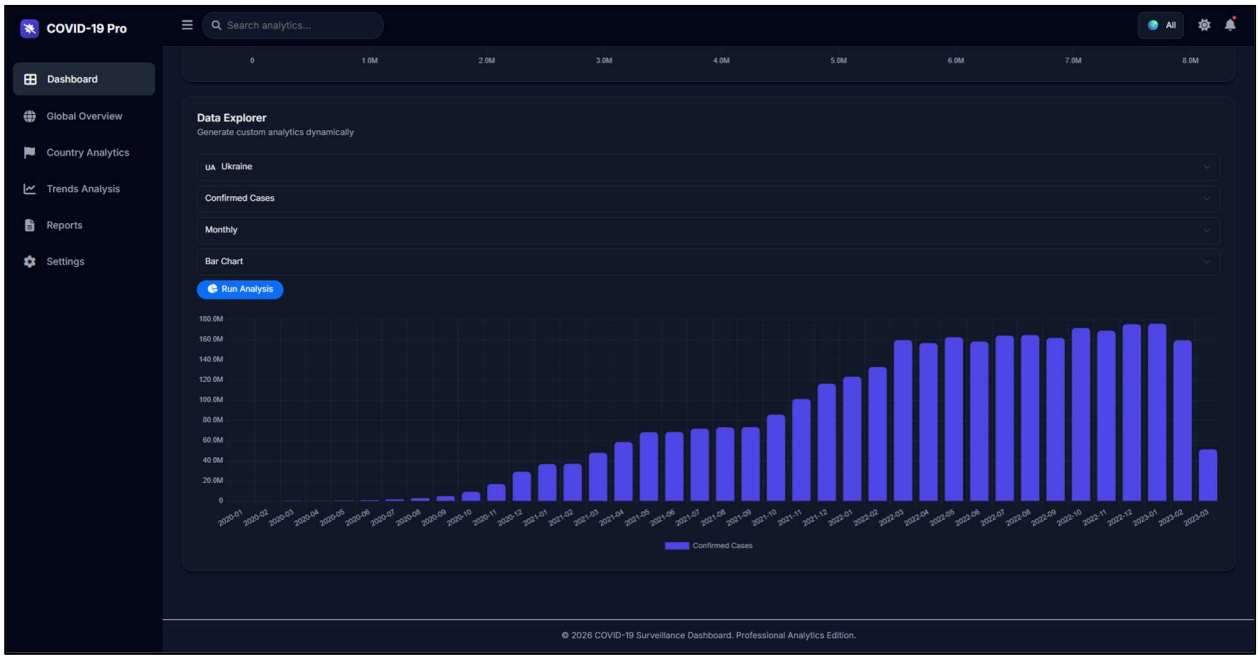
Additionally, the project includes specialized directories like ***graphs/*** for storing exported visualizations. This modular organization improves readability, simplifies debugging, and allows for easy future expansion such as adding new data sources or advanced predictive modules to the platform.

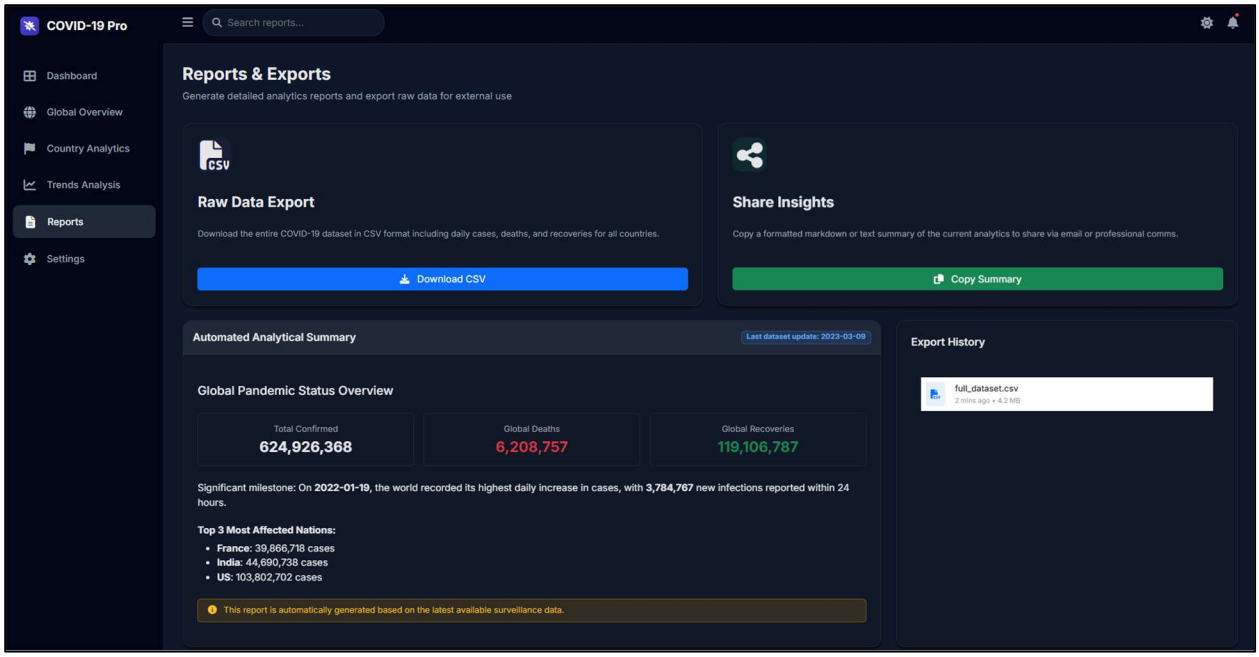
Results and Discussion:

The developed COVID-19 analytics dashboard successfully provides a comprehensive visualization of pandemic data through interactive charts and key performance indicators. The system transforms raw JHU CSSE time-series data into meaningful insights, enabling effective analysis of global and regional health trends.



COVID-19 Pro
Search analytics...
All
Dashboard
Global Overview
Country Analytics
Trends Analysis
Reports
Settings
Daily Growth Rate Volatility
Highest Fatality Ratios per Country (%)
Peru
Mexico
South Africa
Indonesia
Colombia
Hungary
Ukraine
China
Romania
Iran
Global Recovery Trend Over Time
Cases vs Deaths Correlation
Total Deaths
Confirmed Cases


COVID-19 Pro
Search analytics...
All
Dashboard
Global Overview
Country Analytics
Trends Analysis
Reports
Settings
Data Explorer
Generate custom analytics dynamically
UA Ukraine
Confirmed Cases
Monthly
Bar Chart
Run Analysis
Confirmed Cases
© 2026 COVID-19 Surveillance Dashboard. Professional Analytics Edition.



The primary dashboard interface presents critical KPI cards including Total Confirmed Cases, Deaths, Recoveries, and Mortality Rate offering an immediate overview of the pandemic's status. The inclusion of a "Top Affected" indicator helps in quickly identifying the global epicenters of the crisis.

Dynamic Line and Bar Charts provide deep insights into daily infection trends, monthly accumulation, and recovery performance across different nations. The Continent Distribution (doughnut chart) highlights the percentage share of cases globally, while the Mortality Ranking identifies regions with the highest fatality ratios. Specialized visualizations like the Animated Bar Race and Correlation Scatter Plots further reveal the evolving rankings and statistical relationships between cases and deaths.

The dashboard supports robust interactive features, such as searchable country filters and a multi-country comparison tool, enabling users to explore localized data and comparative trends dynamically. The implementation of AJAX-driven real-time updates ensures that visualizations refresh instantly without page reloads, significantly enhancing the user experience.

From a performance standpoint, the system efficiently handles extensive time-series datasets through optimized database indexing and a custom backend caching mechanism. This ensures rapid data retrieval and fluid rendering across all visualization modules. Overall, the results demonstrate that the platform effectively converts complex epidemiological data into clear, actionable public health intelligence.

Conclusion:

The project successfully demonstrates the development of a full-stack data analytics dashboard for COVID-19 surveillance and trend analysis. By integrating automated data preprocessing, indexed database management, a robust RESTful API backend, and dynamic frontend visualizations, the system provides a complete and efficient analytics solution for global health monitoring.

The project highlights the critical importance of rigorous data cleaning and wide-to-long transformation in ensuring the accuracy of epidemiological analysis. It also emphasizes the role of structured database design and strategic indexing in enabling high-performance retrieval of extensive time-series data. The seamless integration of Flask for backend services and Chart.js for interactive visualizations showcases how modern full-stack technologies can be unified to build scalable, data-driven applications.

Through the execution of this project, several key competencies have been developed, including advanced time-series data manipulation using Pandas , optimized MySQL schema design, RESTful API architecture, and professional frontend design with glass morphism techniques. The project also provides practical insights into performance-enhancing techniques such as in-memory caching and asynchronous data fetching.

Future improvements may include the incorporation of Machine Learning algorithms for predictive analytics, such as infection wave forecasting and regional risk assessment. Additional features like vaccination coverage tracking, social mobility correlation, and cloud-native deployment via Docker or AWS can further enhance the system's utility and global scalability.

In conclusion, this project serves as a comprehensive example of how data analytics and full-stack development can be applied to solve real-world public health challenges. It demonstrates the profound potential of data-driven systems in improving pandemic surveillance, assisting in data-backed decision-making, and contributing to informed organizational and governmental responses.